

Supplier Risk & Disruption Cost Analysis

Business Case Study & Financial Analysis

Executive Summary

At-a-glance: ₹612K annual risk → ₹146K Phase 1 savings → 1.4-year payback

This analysis identifies suppliers responsible for **₹612K** in annual supply chain disruption risk. Two high-risk suppliers (Beta Supplies, Delta Logistics) drive 60% of exposure. A phased sourcing intervention can reduce annual risk by **₹146K** (Phase 1, ₹200K investment, 1.4-year payback) and extend to **₹186K+** (Phase 3, with medium-risk supplier optimization).

Business Problem

In a manufacturing operation with production-critical raw materials sourced from 5 external suppliers, delivery delays, quality issues, and inconsistent lead times directly disrupt production. Each disruption incurs significant operational and financial costs. **Current annual disruption risk exposure: ₹612K.**

Core Question

Which suppliers pose the highest disruption risk, and how should we optimize sourcing decisions to reduce financial impact through a phased, data-driven approach?

Analysis Approach (Sub-Questions)

Sub-Question	Analysis Phase	Output
Where is structural vulnerability?	Phase 3: Concentration Risk	Dependency assessment
Which suppliers are unreliable?	Phase 4: Risk Scoring	Risk-ranked supplier list
What is the financial impact?	Phase 5: Expected Loss	Quantified disruption cost
What action should we take?	Phase 6–7: Recommendations	Phased sourcing strategy with ROI

Methodology

Phase 1-2: Performance Metrics

- Evaluated all 5 suppliers on three critical dimensions:
- **On-time Delivery %** (40% weight): Reliability indicator
 - **Defect Rate %** (30% weight): Quality indicator
 - **Lead Time Variability** (30% weight): Consistency indicator

All metrics were normalized (Min-Max scaling) to ensure comparability before applying weighted risk scoring.

Phase 3: Concentration Risk Assessment

Analyzed supplier share per product category. **Finding:** All suppliers represent <50% of any category spend, indicating healthy supplier diversification. **Critical Insight:** Risk is primarily driven by supplier **performance variability** rather than supplier **dependency concentration**.

Phase 4: Risk Scoring Model

Risk Score = 0.4×(1-On-Time%) + 0.3×Defect% + 0.3×Lead_Time_Variability

Scoring Scale: Green (0.0-0.3) | Yellow (0.3-0.7) | Red (0.7-1.0)

Phase 5: Financial Quantification

- Converted risk scores into financial impact:
- Expected Loss = Risk Score × Daily Production Loss × Average Delay Days**
- Daily production loss: ₹50,000 | Average delay: 5 days

Key Findings

Supplier	Risk Score	Risk Level	Expected Loss	Status
Beta Supplies	0.73	HIGH	₹1,83,372	Action Required
Delta Logistics	0.73	HIGH	₹1,81,569	Action Required
Alpha Inc	0.59	MEDIUM	₹1,47,806	Monitor & Plan
Epsilon Group	0.30	LOW	₹75,978	Maintain
Gamma Co	0.09	LOW	₹22,914	Maintain

KEY INSIGHT: Beta Supplies & Delta Logistics account for 60% (₹3,64,941) of total risk.

Financial Impact Analysis

Current vs Proposed State (Phase 1)

Metric	Current	After Phase 1	Impact
Total Annual Risk	₹6,11,641	₹3,13,835	-₹2,97,806 (-49%)
High-Risk Suppliers	2	0	Eliminated
Annual Savings	—	₹1,46,000	+₹1.46L / year
Implementation Cost	—	—	₹2,00,000 (one-time)
Payback Period	—	—	1.4 years

Phase 1 Impact: ₹2,97,806 risk reduction | ₹1,46,000/year savings

Phased Risk Mitigation Strategy

PHASE 1: Immediate Action (Months 0-3) HIGH RISK SUPPLIERS

Action 1: Replace Beta Supplies

- Current: 40% on time, 9.9% defect | Target: 85%+ on time, <2% defect
- Expected Savings: ₹73,372/year | Cost: ₹1,00,000

Action 2: Diversify Delta Logistics

- Current: 46% on time, 14.5% defect | Strategy: Shift 40% to Epsilon Group
- Expected Savings: ₹72,569/year | Cost: ₹1,00,000

PHASE 1 OUTCOME: ₹1,46,000/year savings | Payback: 1.4 years

PHASE 2: Validation & Monitoring (Months 3-6) — PROOF OF CONCEPT

Measure & Validate Phase 1 Results

- Track KPIs: On-time delivery %, defect rate, lead-time variability
- Measure actual savings vs projected (₹1,46,000/year target)
- Document lessons learned from Beta replacement & Delta diversification
- Cost: Monitoring overhead (minimal, ~₹10-20K)

Decision Gate: IF Phase 1 achieves 80%+ of projected savings → PROCEED to Phase 3

PHASE 3: Scale to Medium Risk (Months 6-12) OPTIMIZATION

Apply Proven Strategy to Alpha Inc (Medium Risk Supplier)

- Current Performance: 59% on time (Medium risk, ₹1,47,806 annual loss)
- Strategy: Performance improvement plan + selective diversification
- Investment: ₹60-80K (SLA negotiation, quality audits, training)
- Expected Savings: ₹40-50K/year (if achieves 85%+ on time)

SUCCESS CONDITION:

Only proceed if Phase 1 validates business case.

COMBINED IMPACT (Phase 1 + Phase 3):

- Total Annual Savings: ₹1,86-196K/year
- Combined Investment: ₹2,60-280K
- Combined Payback: 1.5-1.6 years (if both phases succeed)

Phased Financial Summary

Phase	Focus	Investment	Annual Savings	Payback	Decision
Phase 1 (0–3 mo)	High-risk suppliers	₹2,00,000	₹1,46,000	1.4 years	Mandatory
Phase 2 (3–6 mo)	Validation & monitoring	₹10–20K	Data-driven learning	—	Monitoring
Phase 3 (6–12 mo)	Medium-risk supplier	₹60–80K	₹40–50K	1.5–1.6 years (combined)	Conditional (if Phase 1 succeeds)

Business Impact

Dimension	Phase 1 Impact	Phase 1+3 Impact
Financial	• ₹1,46K/year 1.4yr payback	• ₹1,86–196K/year 1.5–1.6yr payback
Operational	40% disruption reduction	50-55% disruption reduction
Strategic	Proof of concept Data-driven approach	Scalable supplier strategy
Resilience	Reduce critical vulnerability	Strengthen overall supply chain

Assumptions & Limitations

- Production Downtime Cost:** ₹50,000/day based on operational impact estimates.
- Phase 1 Improvement Rate:** 40% risk reduction | Range: 30-50%
- Phase 3 Dependency:** Only proceed if Phase 1 validates 80%+ of projected savings
- Concentration Risk:** Simulated from spend distribution
- Lead Time Variability:** ~30 orders/supplier (may not capture seasonality)

Implementation Timeline

Phase	Duration	Cost	Outcome
Phase 1: Qualification	4 weeks	₹50K	Supplier qualified
Phase 1: Transition	4 weeks	₹50K	New suppliers active
Phase 1: Stabilization	4 weeks	₹1,00K	Performance validated
Phase 2: Monitoring	12 weeks	₹10–20K	Measure results
Phase 3: Implementation	24 weeks	₹60–80K	Medium risk optimized

PHASE 1 TOTAL: ₹2,00,000 | PHASE 1+3 TOTAL: ₹2,60-280K

Conclusion

This phased approach quantifies ₹6,11,641 in annual supply chain disruption risk and proposes a scalable, data-driven mitigation strategy. Phase 1 targets high-risk suppliers (Beta, Delta) with ₹2,00,000 investment achieving ₹1,46,000 annual savings (1.4-year payback). Phase 2 validates results through rigorous monitoring. Phase 3 extends the proven strategy to medium-risk supplier (Alpha Inc) for combined ₹1,86-196K/year benefit.

This phased approach de-risks implementation, proves business case, and enables scaling to medium-risk suppliers only after validation of Phase 1 success.

This model enables proactive, data-driven supplier risk management with measurable financial impact.

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Analysis Method: Risk Score Model + Phased Financial Impact Modeling

Data Source: Procurement KPI Dataset | 777 records | 5 suppliers